Habitat Selection and Home-Range Estimation

Continuous-Time Animal Movement Modeling

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Ray Rui, 2018

Overview

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1) Resource selection functions (RSFs)
2) (Autocorrelated) Kernel density estimation
3) (A)KDE discontinuity bias
4) RSF-informed (A)KDE

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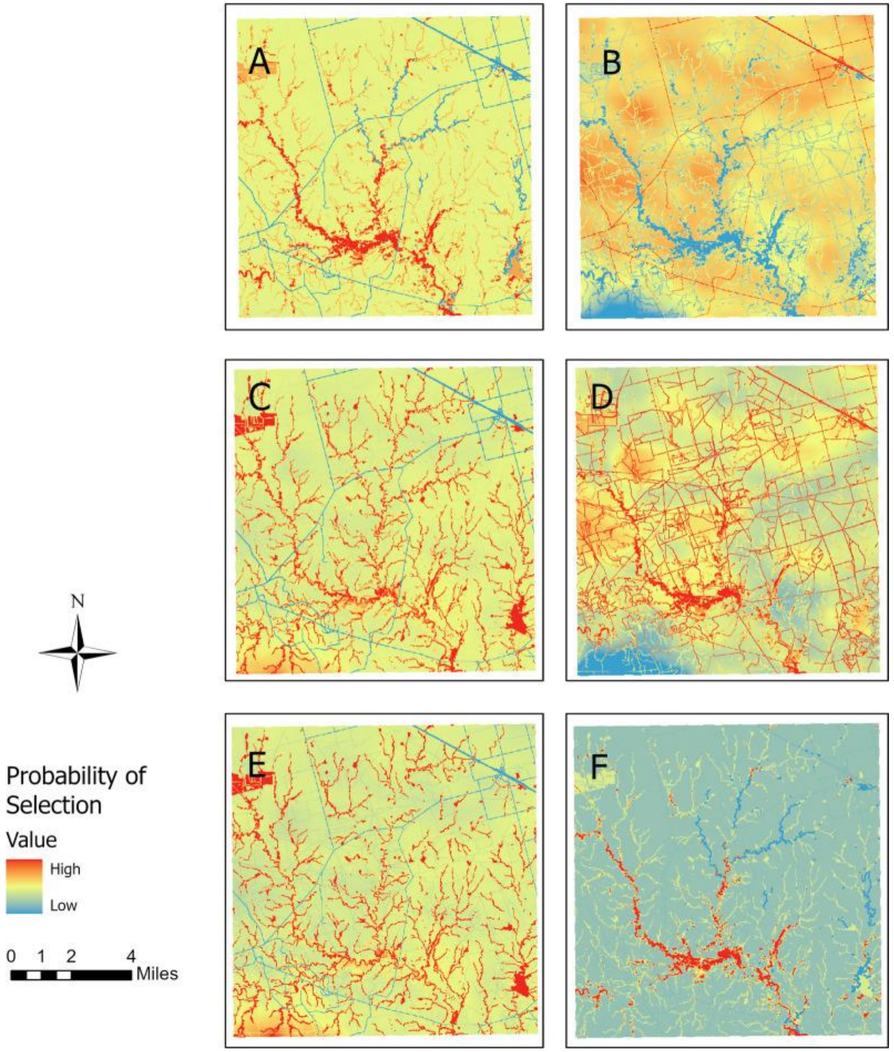
Resource Selection Functions (RSFs)



We are often interested in **habitat selection** and what drives animal movement.

- How do available resources impact an individual's home range?
- What environmental factors does an animal prefer or avoid?
  - How do we quantify the relationships between animal locations and covariates?
  - How might this change over time?
  - How does human disturbance on a landscape affect an animal's space use?

Yang *et al.* (2023)

Resource Selection Functions (RSFs)

RSFs are usually framed as logistic regression models that compare **data** to random **available** points.

$$p(x, y) \propto \exp(\beta_1 R_1(x, y) + \beta_2 R_2(x, y) + \beta_3 R_3(x, y) + \cdots)$$

Where w denotes relative preference, positive β values indicate preference for the corresponding covariate, and negative β values indicate avoidance.

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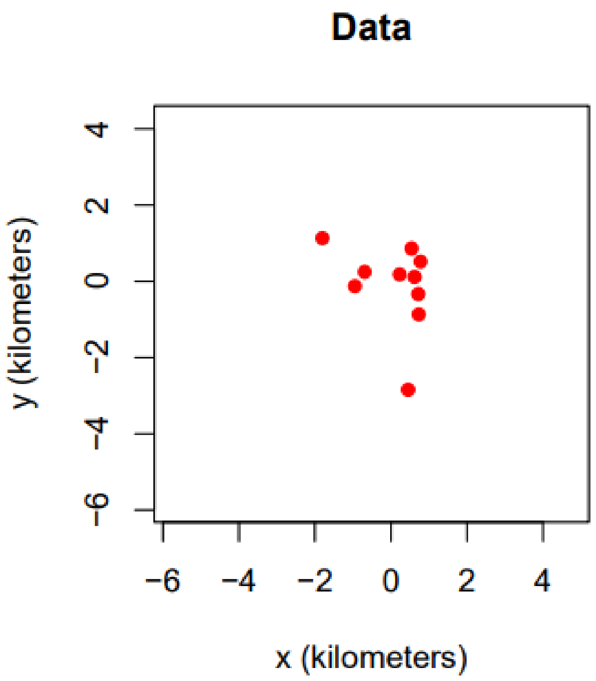

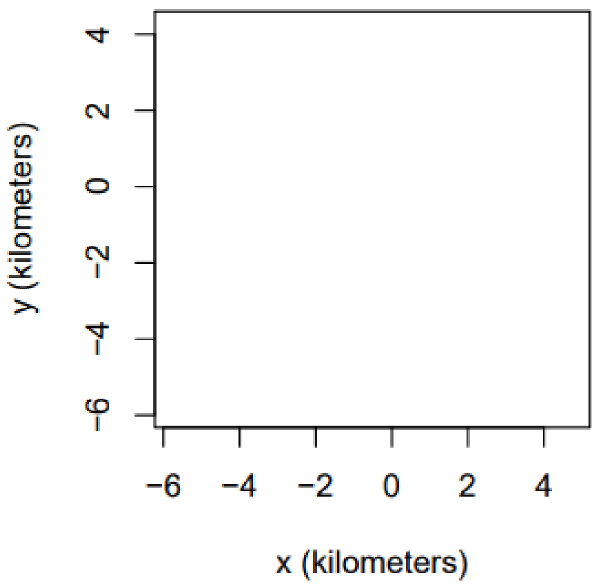

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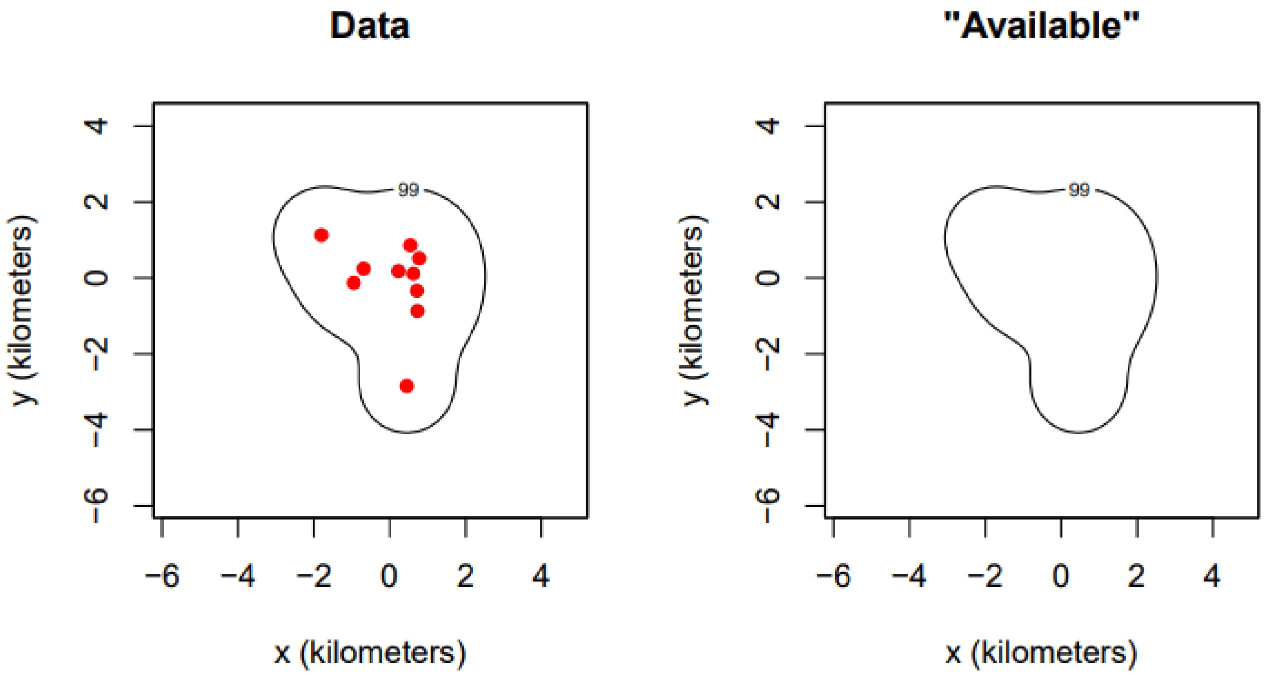

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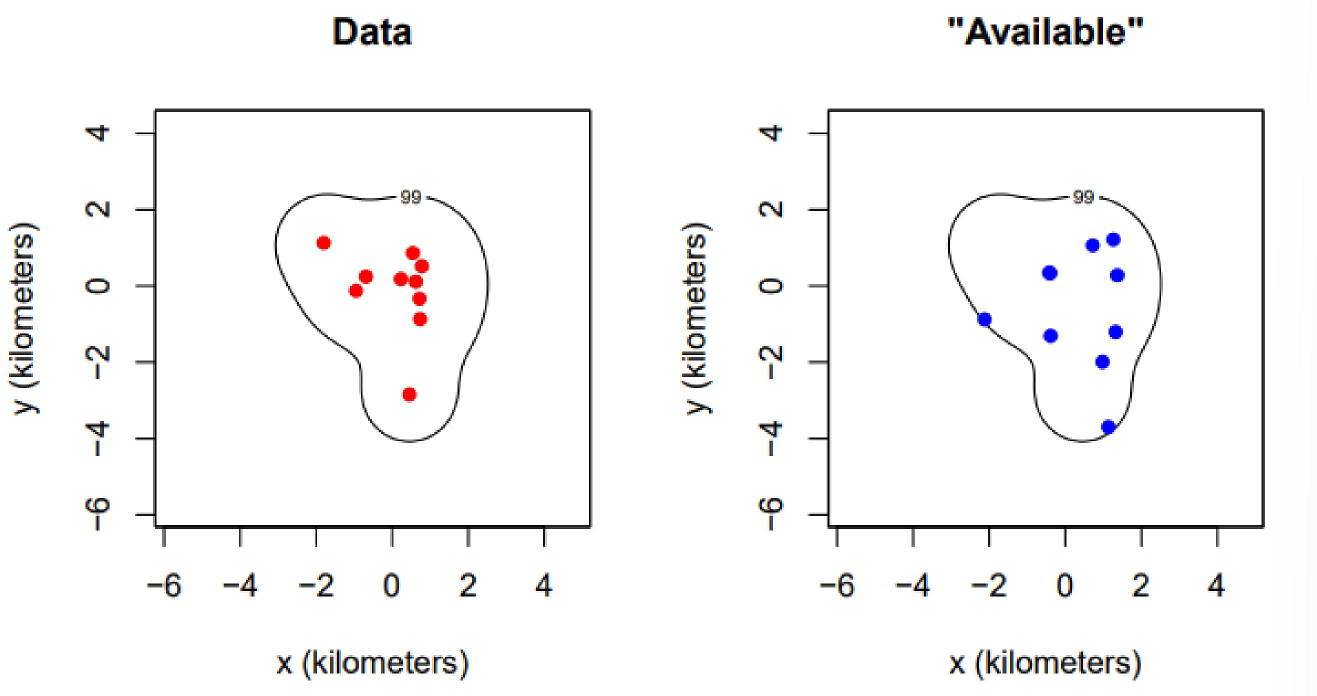

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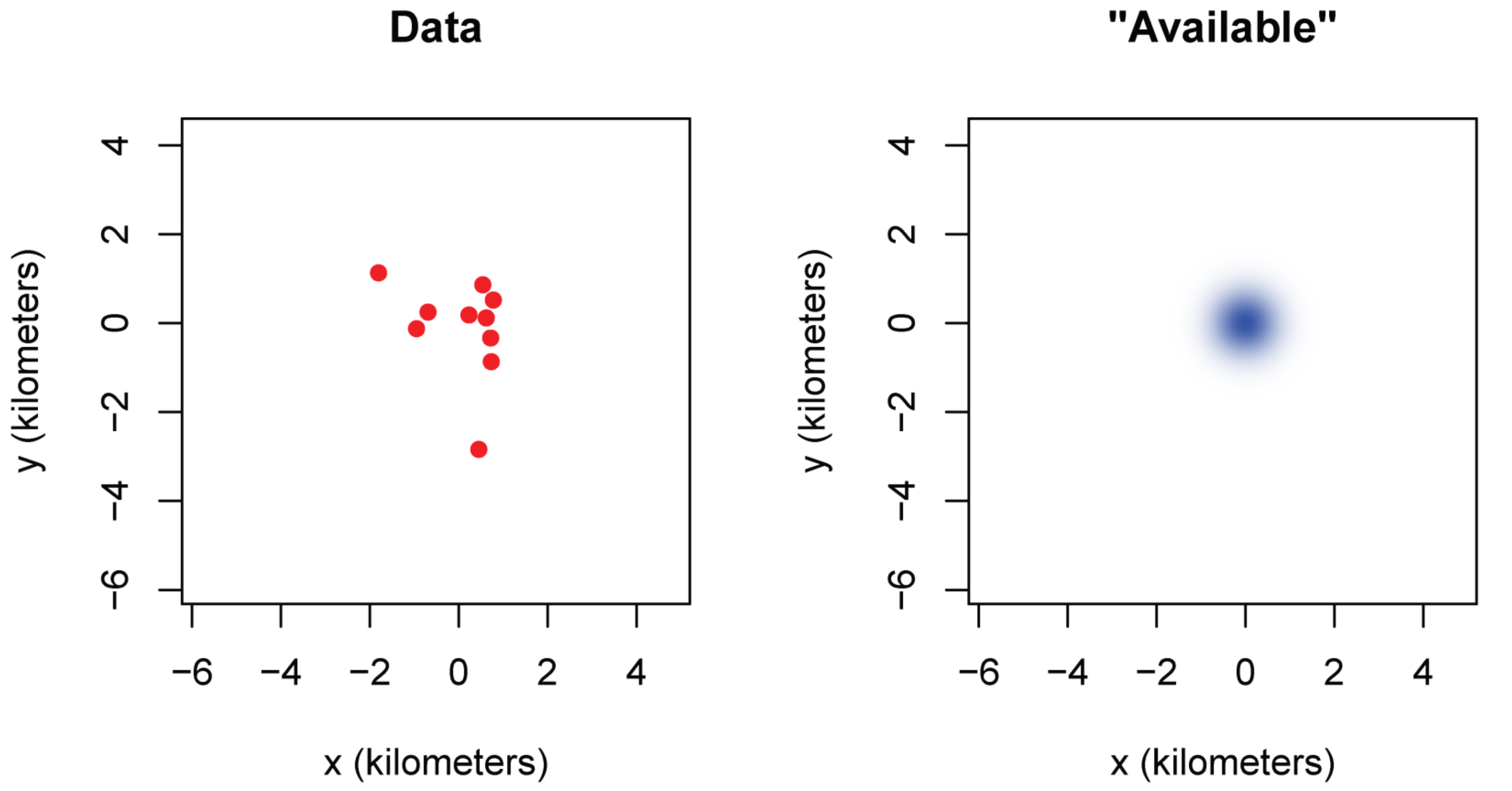
- Also accounts for temporal autocorrelation, sampling irregularity, and the appropriate number of available points to sample

Integrated Resource Selection Functions (iRSFs)

iRSFs estimate the scale of the "available" area simultaneously with the resource selection parameters.



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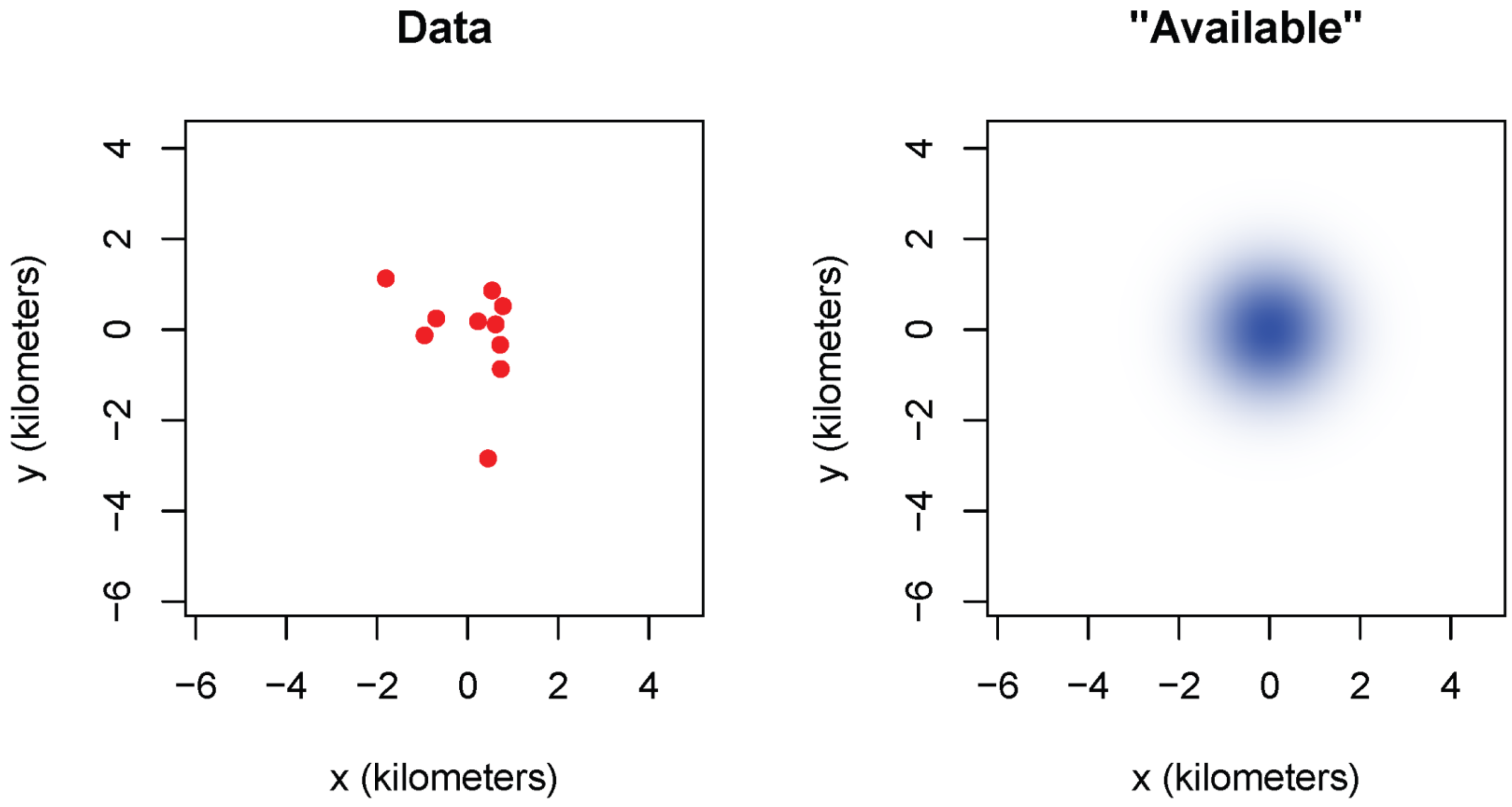

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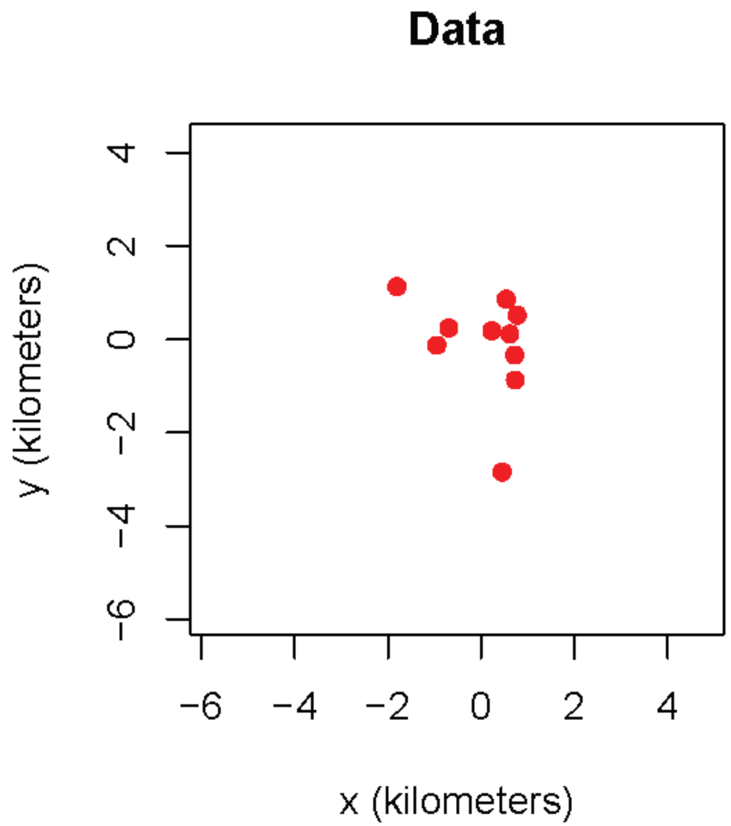

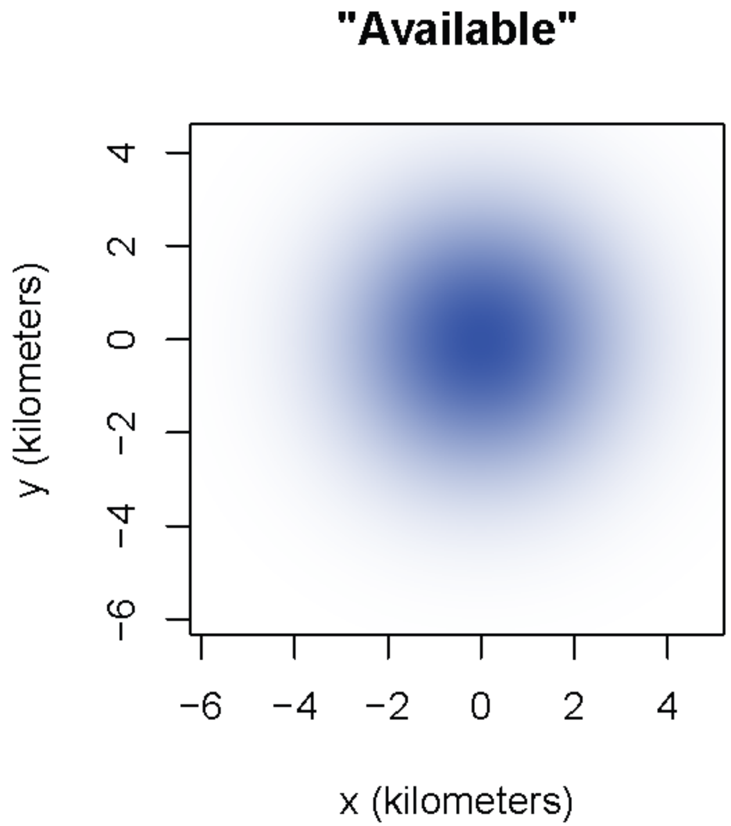

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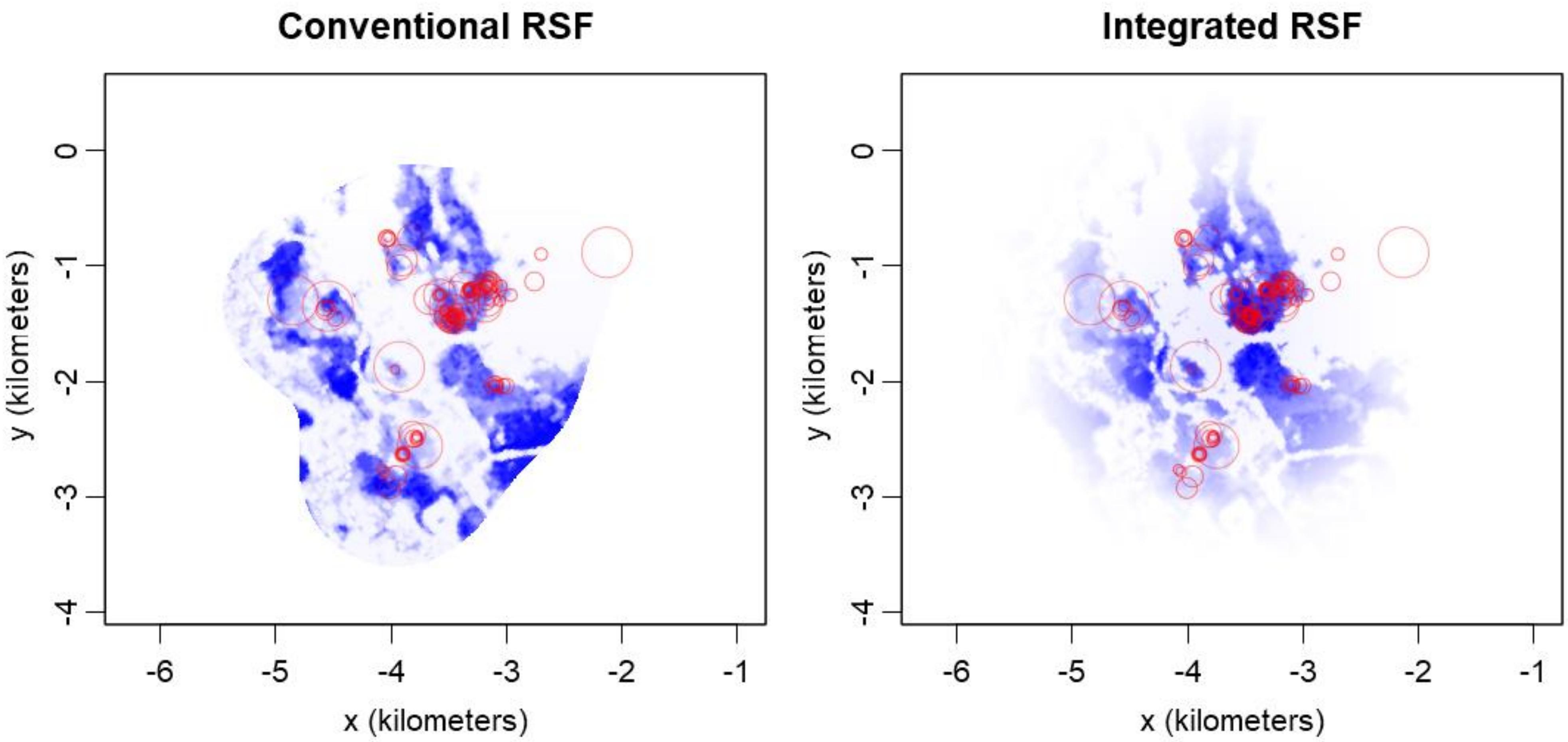
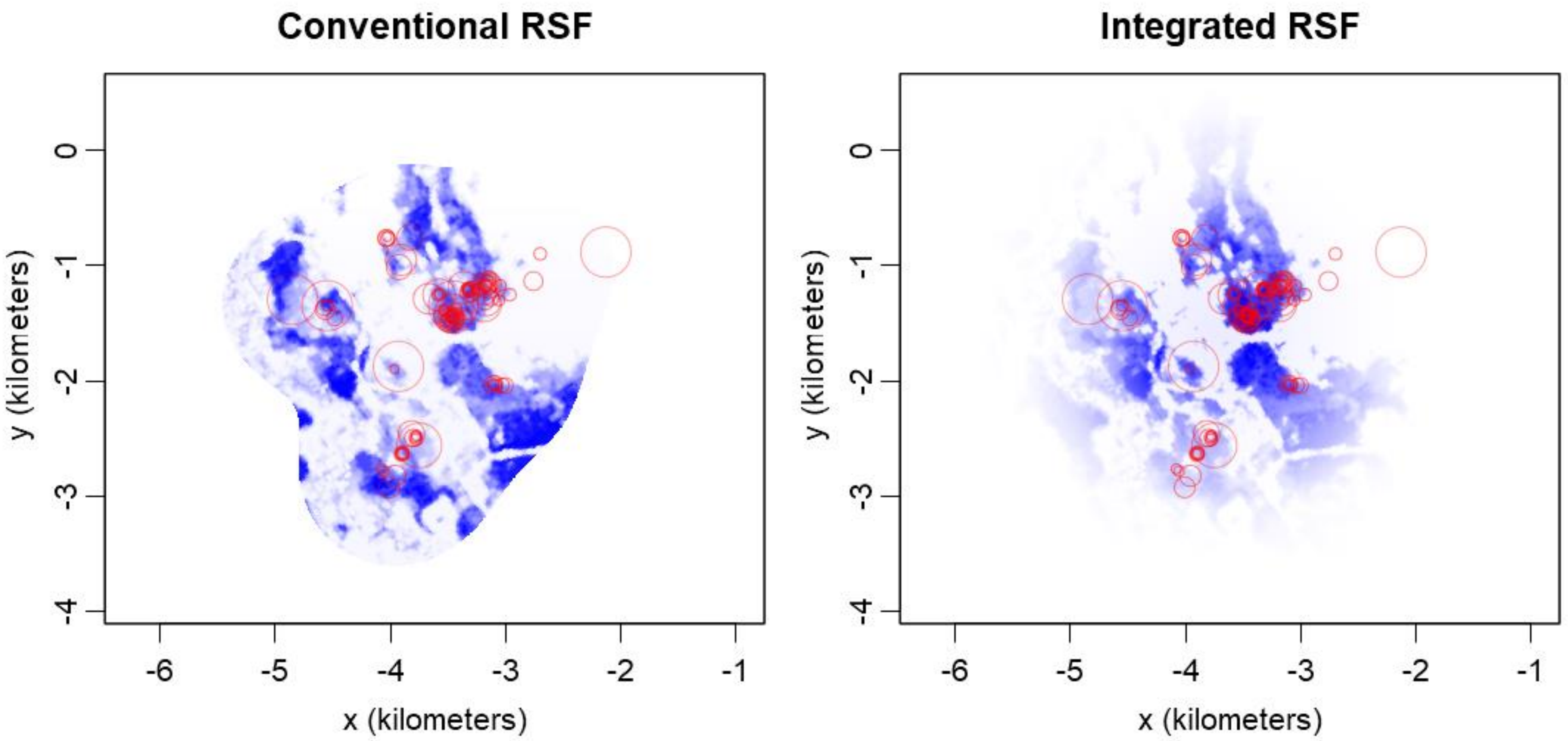
- Conventional RSFs do not estimate the most likely "available" area.
- Resource selection coefficients can be sensitive to the "available" area parameter, in which case conventional RSFs do not estimate the most likely resource selection coefficients (bias).

Conventional versus Integrated RSFs



Conventional versus Integrated RSFs

The conventional estimate fails to cross validate any location outside of the "available" area.



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- But it might not be a very good one…
- And it probably isn’t asymptotically consistent

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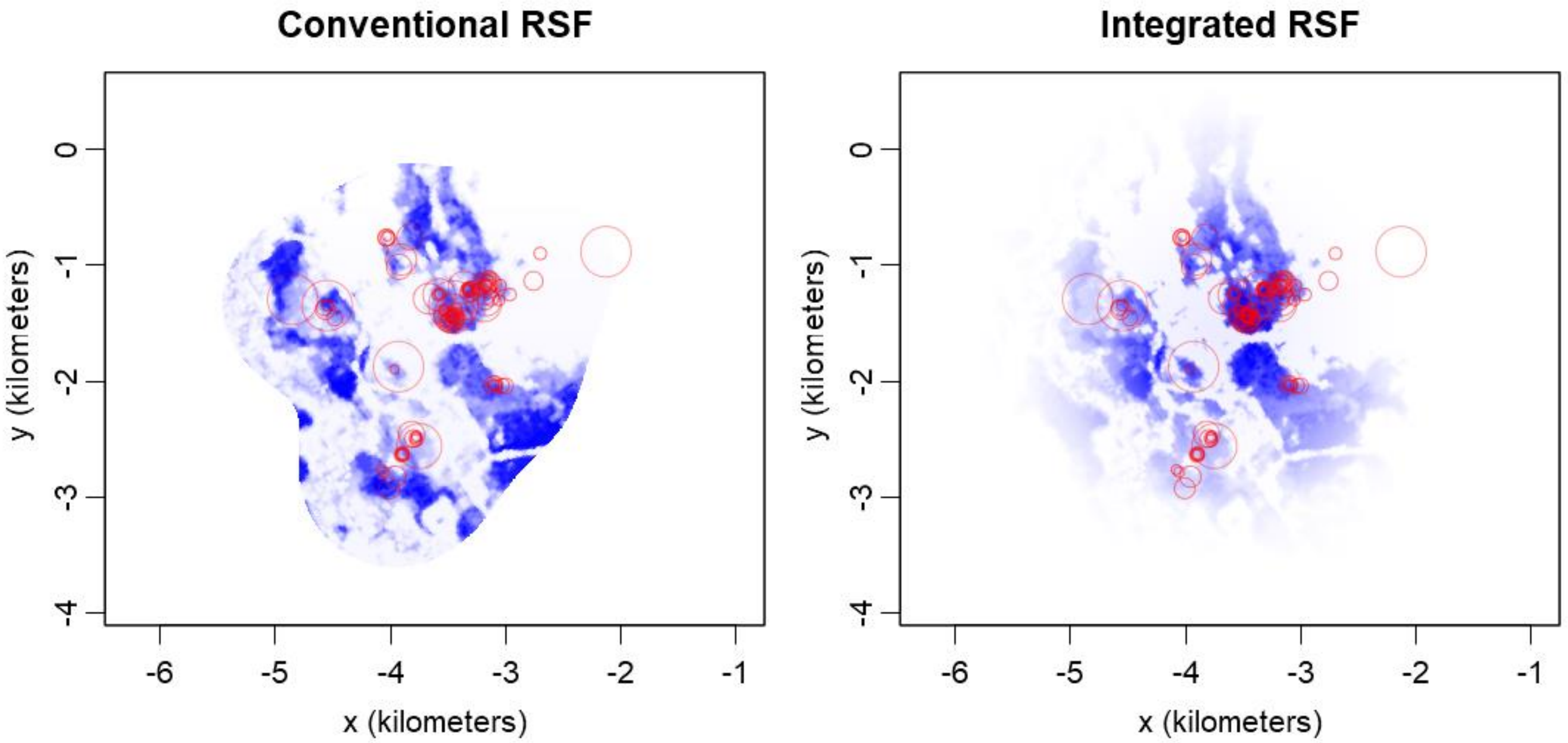

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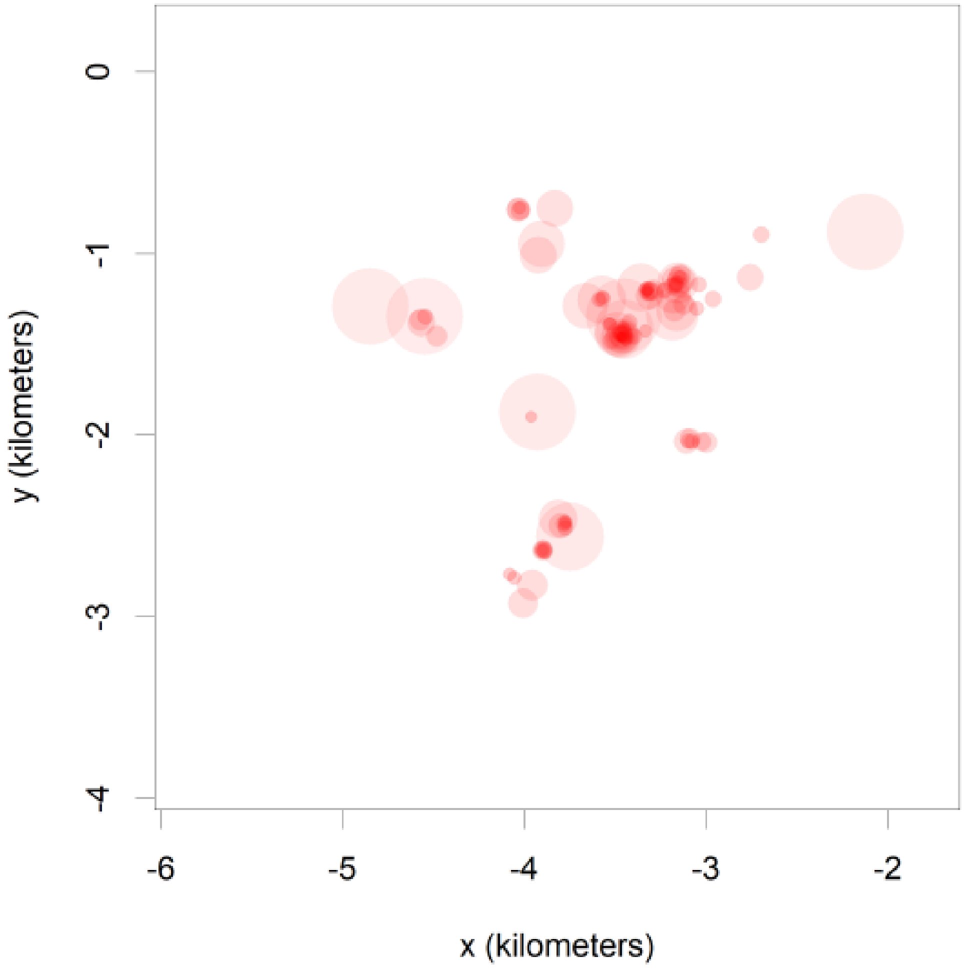

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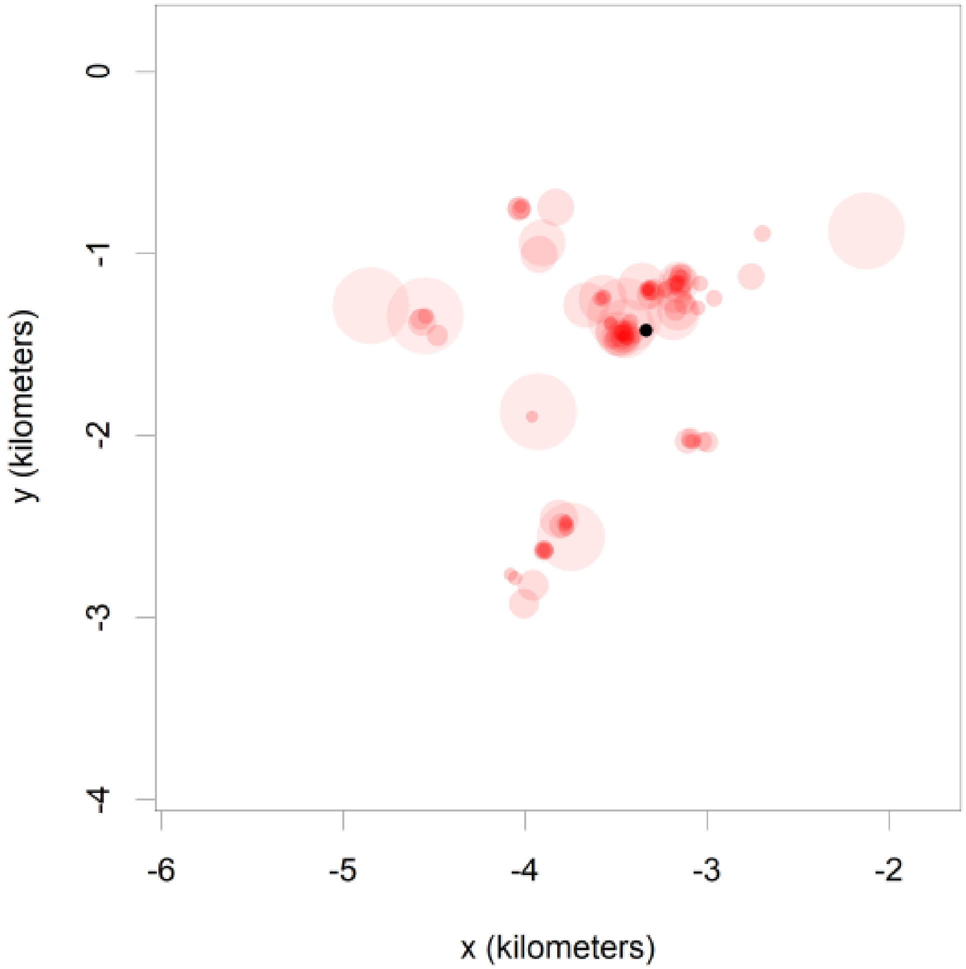

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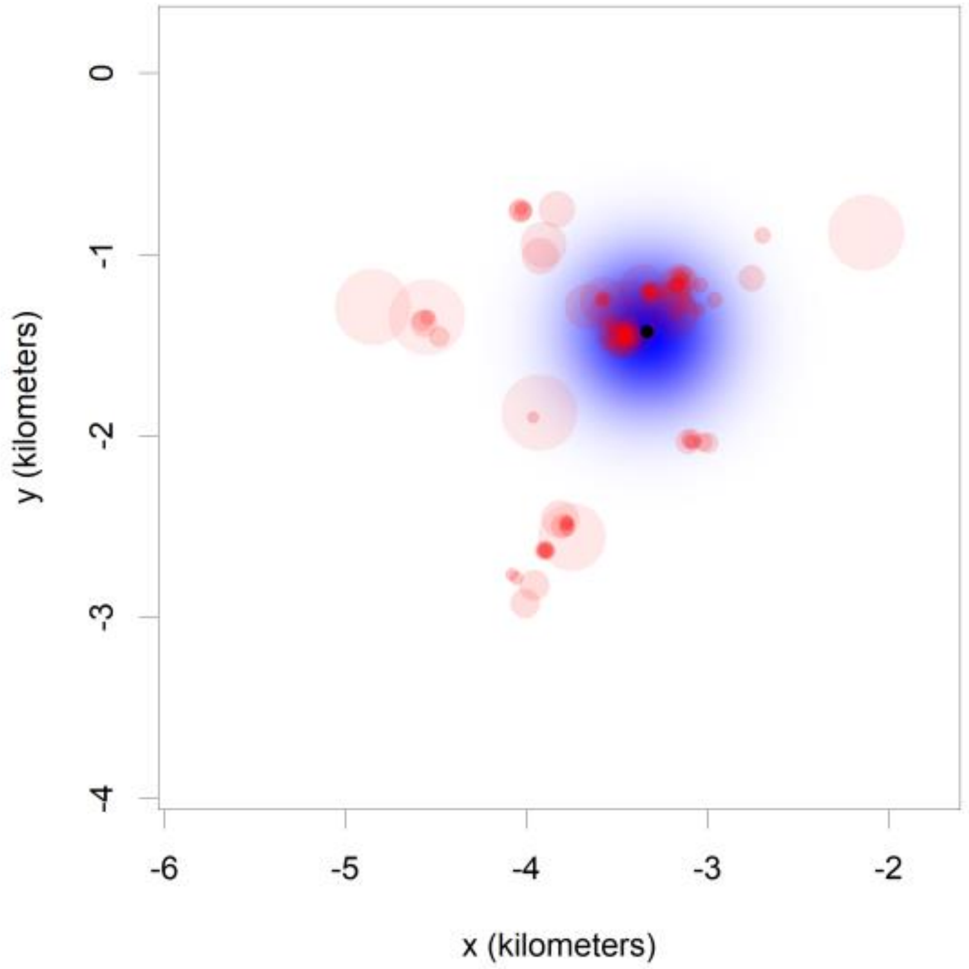

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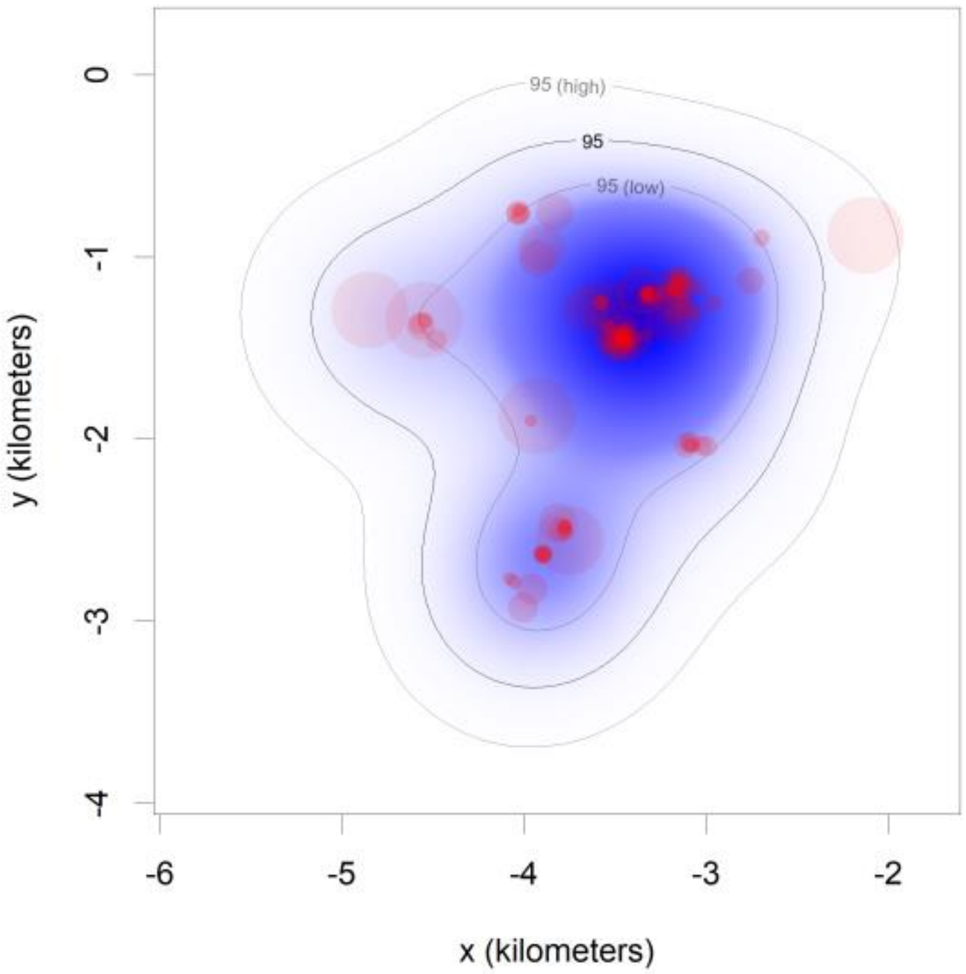

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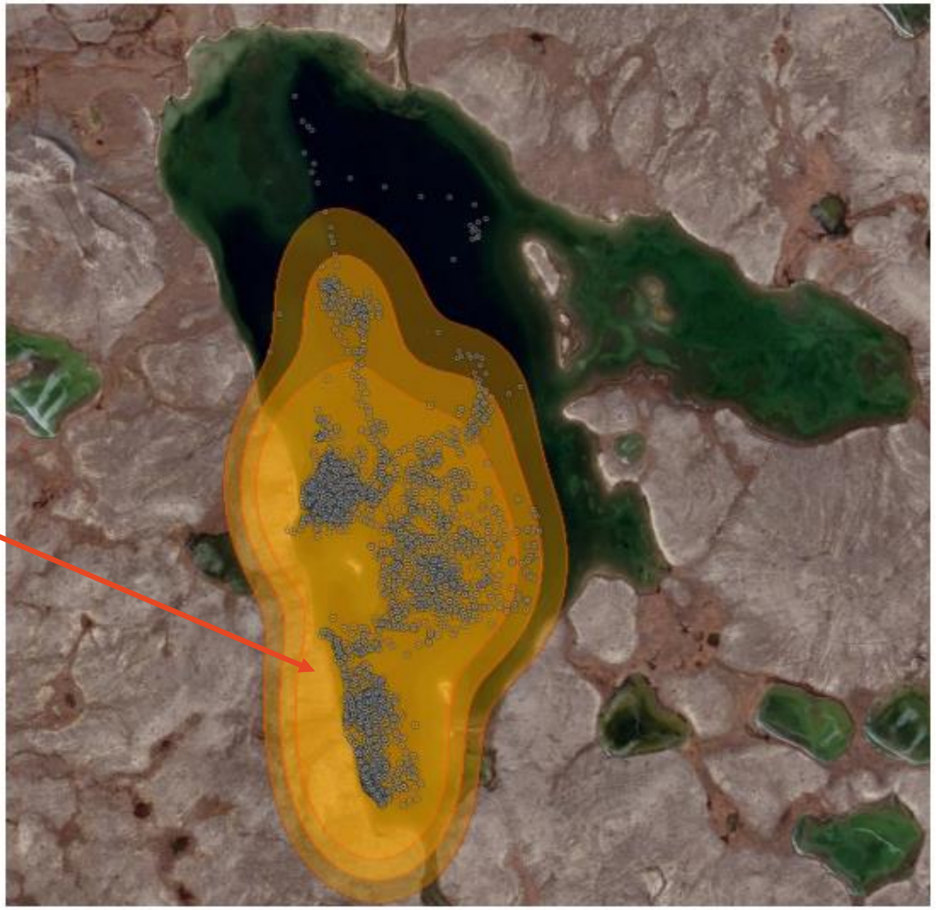
Habitat edge



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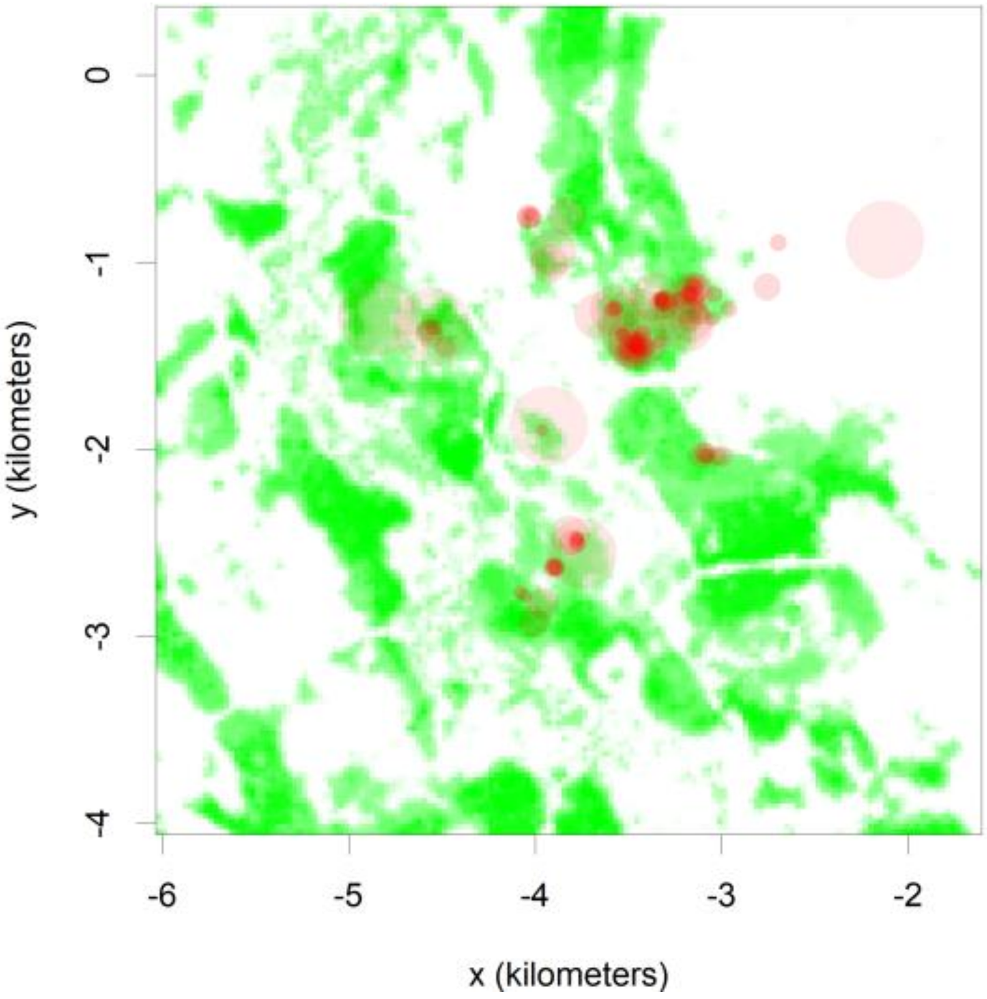
- Now coded up in the `ctmm` R package

RSF-informed (A)KDE



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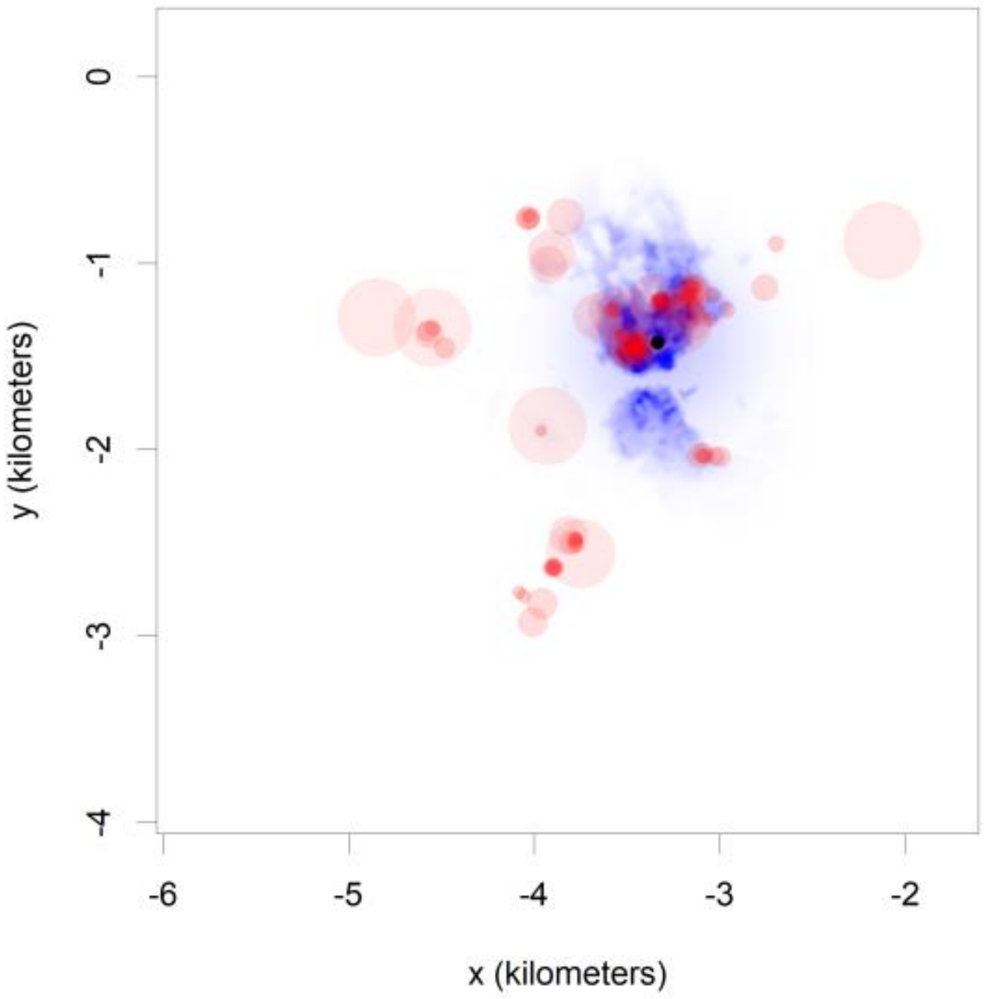

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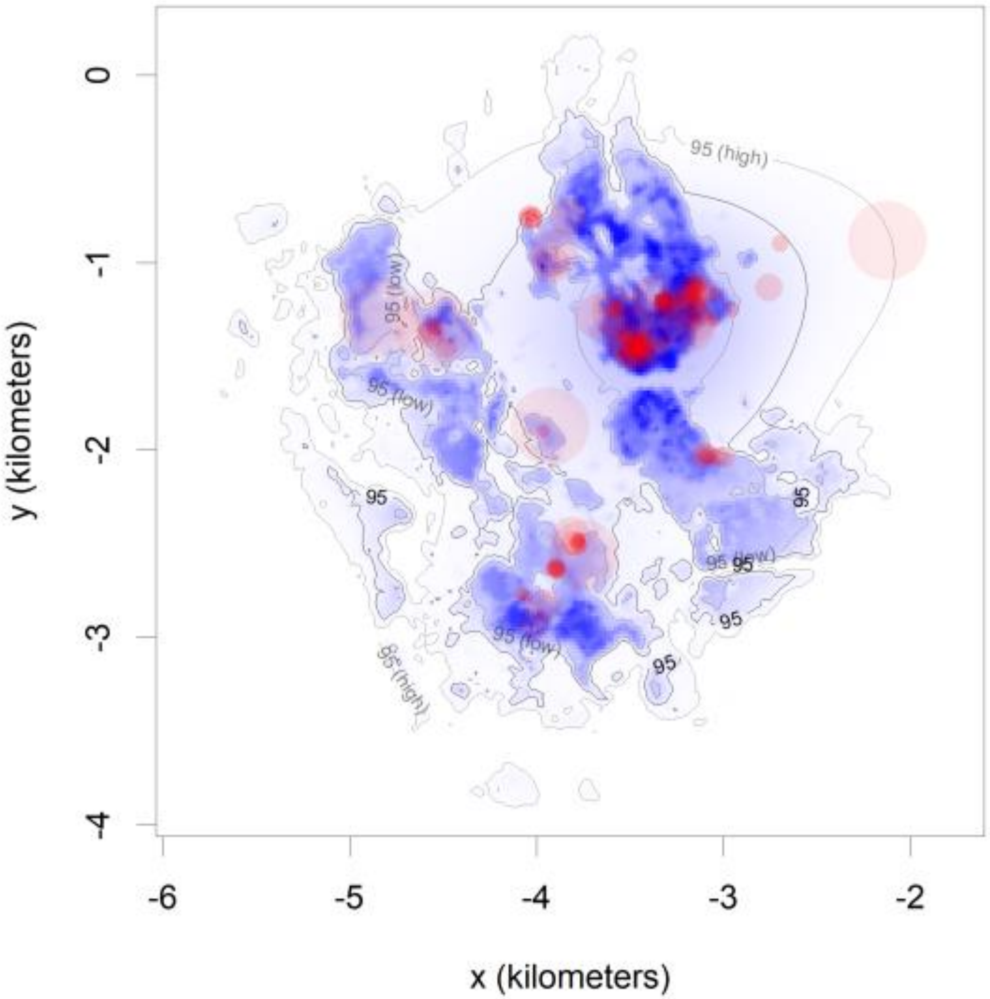
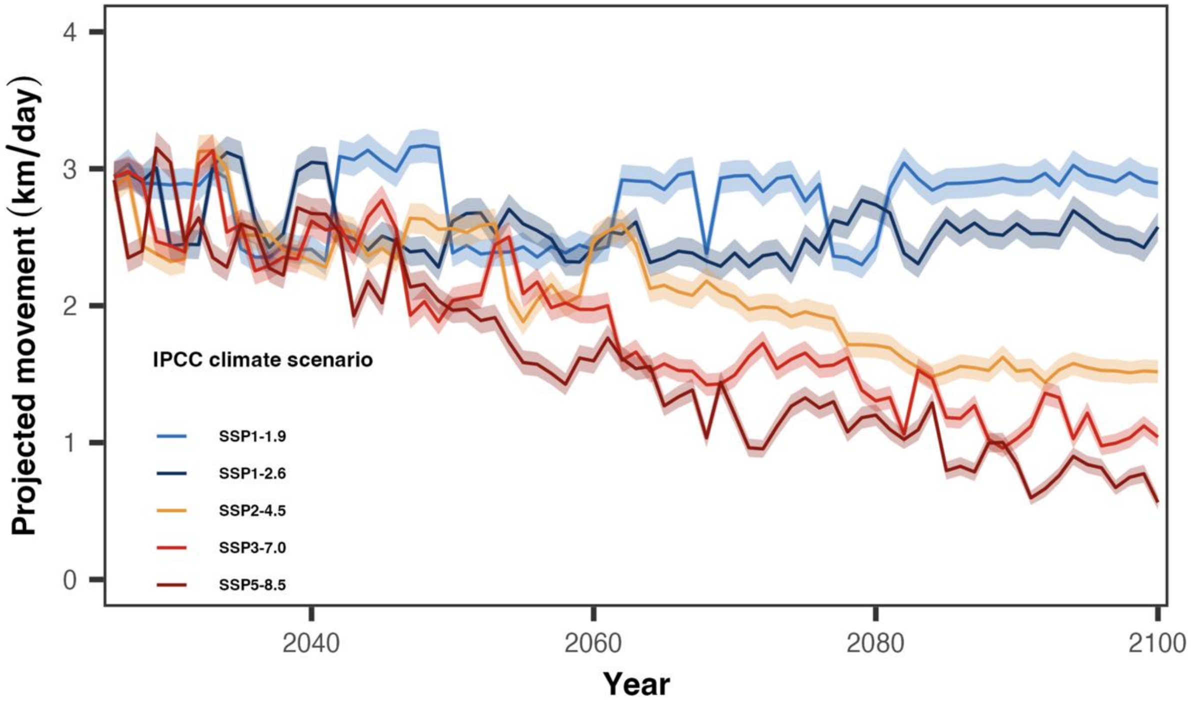


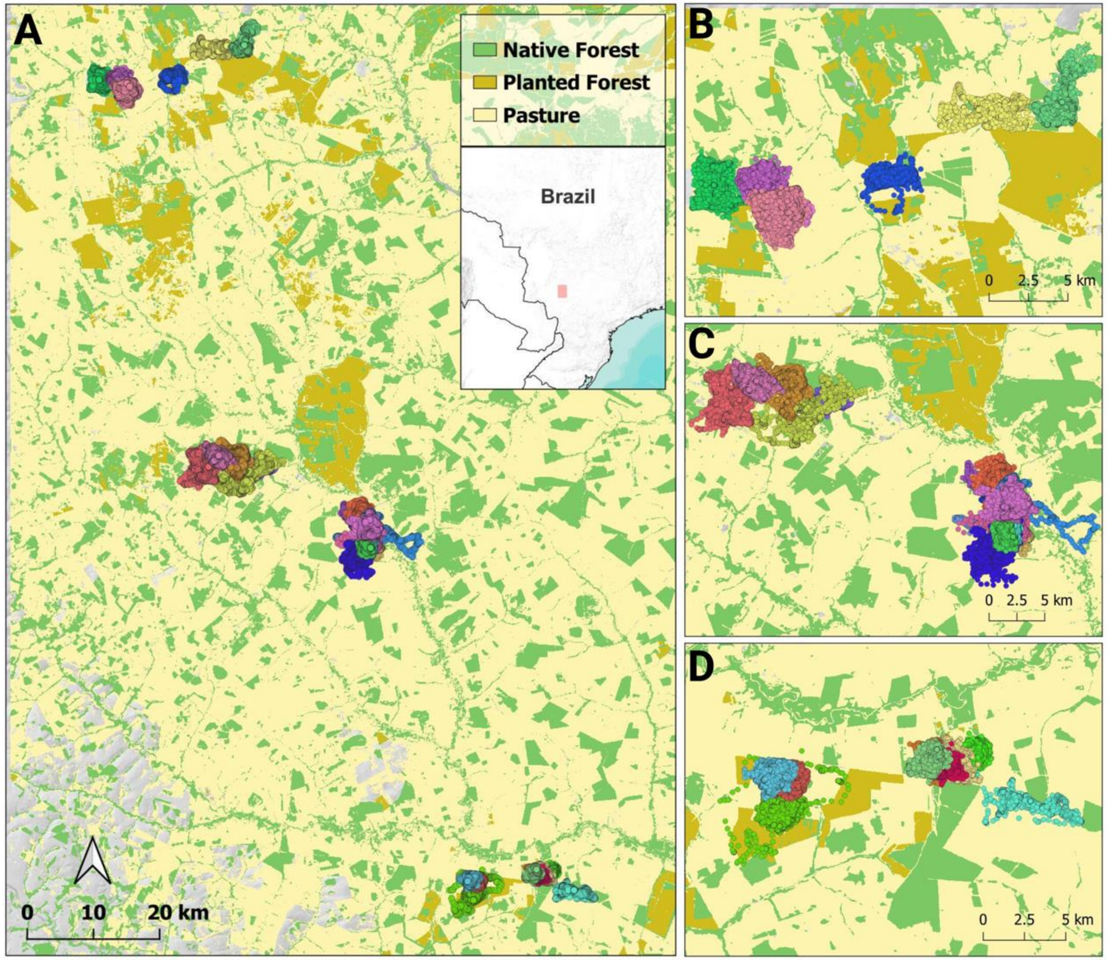
Applications

Mascaretti, 2023

Giant Anteater
(*Myrmecophaga tridactyla*)



Noonan *et al*. (2026)



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4) What's missing / future work:
   Diffusive kernels that respect movement

ctmm_rsf.R

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Conservation Biology Institute

CASUS
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